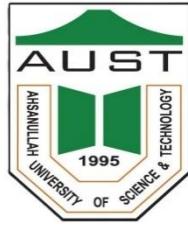


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Department of Computer Science and Engineering
Program: Bachelor of Science in Computer Science and Engineering

Project Title: Mini Craft

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Introduction

1.1 Project Overview

Mini Craft is a 2D C++ game built with the iGraphics/GLUT framework in Visual Studio 2013. It combines an exploration map, a persistent home-base hub with resource gathering and crafting, and a wave-based battle system, tied together by a single game-state machine

1.2 Objective

- To develop an interactive 2D game using the iGraphics library.
- To implement a resource-gathering and crafting economy (stone/wood/iron/water → potions, armor).
- To implement a wave-scaling battle system with regular boss encounters and a persistent save/load system.

Game Development

2.1 Tools & Technologies Used

- **Programming Language:** C, C++
- **IDE:** Visual Studio 2013
- **Graphics Library:** iGraphics (custom wrapper) over GLUT/OpenGL
- **Platform:** Windows PC
- **Version control:** Git/GitHub

2.2 Core Features

- **Character selection:** Guardian, Ranger, Alchemist fighters
- **Home base hub:** Movement (WASD/arrows), attack, class-specific ability (Guardian shield / Ranger dash)
- **Resource gathering system** — 4 node types (Stone, Wood, Iron, Water), each with distinct gather
- **Crafting system** — Potions (Movement, Power, Reinforcement, Health) and a Damage Scroll
- **Armor crafting** (Iron/Rusty/Steel Armor): via a separate panel using the same UI pattern
- **Boss battles** every 5th level (auto-triggered), plus an on-demand Boss button after a full wave clear

Implementation Details

3.1 Code Structure

The game consists of multiple modules:

- **iMain.cpp** – Initializes the game, starts the iGraphics engine
- **Menu.h** -Main menu UI
- **Player.hpp** – Player stats
- **Fighters.hpp** – Guardian/Ranger/Alchemist definitions
- **Enemy.hpp** -Enemy definitions used in battle
- **Battle.hpp** – Wave based battle logic
- **Homebase.hpp** – Home base hub logic
- **HomeBaseConfig.hpp** – Home base constants/config
- **GatherSystem.hpp** – Resource node gathering logic
- **CraftingEngine.hpp** – Crafting book UI and crafting logic
- **ArmorEngine.hpp** – Armor crafting panel
- **PlayerBuffs.hpp** – Effects of consumed crafted items
- **Inventory.hpp** -Inventory tracking
- **SaveGame.hpp** – Saves the stats of the game

3.2 Challenges Faced & Solutions**

Challenge	Solution
Implementing potion logic	Used a variable in all the character specification
Boss fight movement and position	Used a random coordinate for both the Boss and Hero
Hero/Enemy Sprite Graphics	Use of better prompt in Gemini

Screenshots

Figure 1: MENU



Figure 2: Character Selection



SELECT CHARACTER



The Ranger

Shadow Dasher

A master of speed and precision, the Ranger strikes from the shadows and vanishes in an instant. With unmatched range and agility, he controls the battlefield with ease.



Hitpoints:
70

Damage per second:
94

Range:
500 Tiles

Damage type:
Single Target

Targets:
Ground & Air

Favorite target:
Any

Housing space:
5

Movement speed:
320

Special Ability



Shadow Dash

Dash forward quickly, evading attacks and outmaneuvering enemies.

Dash Distance: 150 Tiles • Energy Cost: 25



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SELECT CHARACTER



The Alchemist

Witchcraft Expert

A master of ancient lore and mystical brewing, The Alchemist is a true expert of witchcraft. Feared for her deadly concoctions and revered for her life-giving elixirs, she controls the flow of life and death on the battlefield. With an unparalleled understanding of magical elements and a quick hand with potions, she supports her allies and decimates her foes with ease.



Hitpoints:
100

Damage per second:
94

Range:
450 Tiles

Damage type:
Single Target

Targets:
Ground & Air

Favorite target:
Any

Housing space:
5

Movement speed:
200

Special Ability



Potion Proliferation

Quickly mixes and distributes potions, reducing the resource cost for allies and enabling faster potion use for herself.

Resource Cost: -25% • Potion Consume Speed: x1.5



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Figure 3: Skeleton Fight



Figure 4: Zombie Fight



FIGURE 5: BOSS Fight



Figure 6: Inventory



Figure 7: Potion Crafting



Figure 8: Armor Crafting



4. GitHub Collaboration

4.1 Repository Management

This project was handled using Git and GitHub. Members divided the work between them and Used GitHub to merge it. This kept everyone's work isolated during development and avoided disrupting the main branch until the work was ready to be combined. Merging the code together using Git was found to be a bit difficult for the members so there may be a slight miscalculation of Members' individual Contribution graph

4.2 Contribution Workflow

The team followed a structured workflow:

- Tasks were divided among the three team members at the start.
- Each member worked independently on their own branch (main_nur, main_maria, main_raya) on their assigned share of the work.
- Branches were kept separate and only merged into main shortly before checkpoint deadlines.
- This was deliberately isolating work until merge time avoided mishaps on main that would be a hassle to resolve mid-development.

4.3 Individual Contributions

Each member was responsible for specific modules:

- **MARIA AKTER:** Built core gameplay systems: resource interactions, hero special abilities, inventory/login/equip logic and visual effects.
- **NUR ZAMAN LAM:** Built the main menu UI and character selection, plus hero sprites then moved into equipment/crafting mechanics and boss fight logic.
- **ADRITA TASNEEM RAYA:** Handled the tutorial flow and "Save the Witch" cutscene/event logic, background and cutscene art, hero sprites, plus armor art and save-game functionality.

4.4 Contribution Evidence

GitHub Contribution Graph

N.B: The team faced some issues committing to GitHub throughout the Checkpoints as a result the members had to push the whole project directly before the Third Checkpoint. After that debugging was done to make the game function smoothly and the members were able to commit. So, the contribution graph may not have enough information to calculate the correct contribution rate for each member



Challenges & Limitations

5.1 Challenges

One significant challenge was merging the code of the teammates in the main project that led to several errors and needed quite some time to fix. Also generating the correct movement sprite for the characters was challenging.

To address these obstacles, the team made use of AI tools during development:

- **Gemini:** Used to generate the necessary graphics for the game.
- **Claude:** Used to help write some of the more complex sections of code and to debug.

Conclusion & Future Improvements

6.1 Summary

Mini Craft delivers a full gameplay loop — menu, character selection, a story map, and a persistent home base with gathering, crafting, and combat — built on iGraphics with a single unified state machine and a working save system.

6.2 Future Enhancements

- Additional boss types beyond the current Dragon
- Expand crafting tree / more armor tiers